

Crispr-Predictor: A Machine Learning Model for OFF-Target Prediction in CRISPR-Cas9

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Abstract—The CRISPR-Cas9 technology enables incredibly accurate genome editing however, its biomedical application is often constrained by unintended off-target modifications. In addition, these unintended modifications cause the degradation of precision and reduce biosafety and specificity, the case needs to be made that in-silico approaches be used for predicting off-target activities. In this way, the design of guide RNA can be more efficient, more protected, and safe. Recent investigations have shown that when using Deep Learning (DL) methodologies, it helps model sequence–structure relationships and therefore predicts off-target activities. This work adds a data-driven predictive tool based on deep learning techniques analyzing sequence-based features.

One architecture, i.e., CNNs and the other, RNNs is developed and implemented in two different configurations for the analysis. Each architecture is implemented with one-hot encoding and k-mer-based sequence representations to model various sequence dependencies. This framework automates the entire stages such as preparing the dataset, transforming the sequence, training the model, and verifying the performance via accuracy, precision, recall, F1-score, AUC-ROC, AUC-PR, sensitivity, and specificity. Experimental results on datasets with mismatches and insertion–deletion mutations also show that the right mix of encodings paired with deep learning architectures substantially improves the prediction capability. Compared between two models, one shows that when encoding methods are combined into CNN and RNN architecture, they bring better stability and accuracy compared to the base models. This framework, therefore, can be applied as an efficient and flexible computational tool to make CRISPR guide RNA design safer and precise.

Index Terms—CRISPR-Cas9, Off-target Prediction, Deep Learning, CNN, RNN, LSTM, One-hot Encoding, K-mer Encoding, Genome Editing, Guide RNA Design, Bioinformatics, Computational Biology.

I. INTRODUCTION

The Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-Cas9 technology is one of the most promising and transformative tools for genome editing. The unprecedented ability to introduce targeted double-strand breaks in DNA with extremely high precision has helped drive substan-

tial research in genetic therapy, crop improvement, and disease modeling [1],[2].

Although its efficacy is remarkable, an inherent limitation for both clinical and therapeutic use of CRISPR-Cas9 is the ability to generate off-target effects—that is, modifications to cells at genomic sites that are not intended to elicit targeted changes in the genome yet which bear some similarity to the targeted CRISPR target [3], [4]. The off-target changes can lead to genomic loss, the induction of unwanted mutations, and a significant safety risk in therapeutic settings. Accordingly, to achieve trustworthy and robust applications, CRISPR-based methods need to be built to predict and/or downregulate off-target activities.

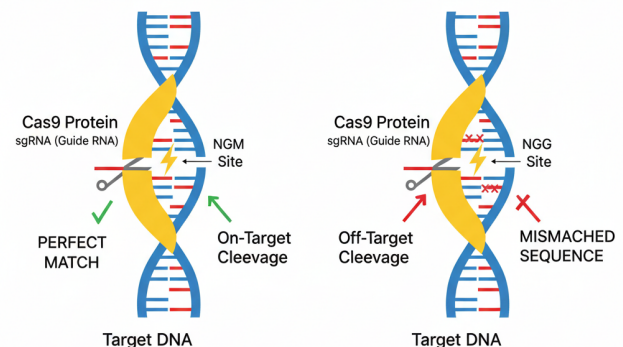


Fig. 1: Mechanism of Crispr/Cas9

Initial computational approaches for CRISPR off-target prediction largely utilized alignment-based scoring, thermodynamic models, or empirical mismatch counting as a metric to assess the possibility of off-target effects [5]. Although these were significant methods, their predictive effectiveness was hampered by their failure to model nonlinear dependencies in DNA sequence settings. Machine learning and deep learning can be seen to bring a drastic paradigm shift that allows the prediction of complex sequence–function relationships from data and has provided tools for learning these relationships directly from large data sets [6], [7], [8].

Deep learning architectures, e.g., Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have been increasingly successful in modeling local motif associations and long-range dependencies in nucleotide sequences respectively [9]. This study introduces a novel hybrid deep learning model, which merges CNNs and RNNs, and performs CRISPR-Cas9 off-target activity prediction.

The CNN component extracts local sequence motifs using convolutional filters that capture spatial relationships with nucleotide pairings by identifying spatial relationships in nucleotide arrangements. The RNN component, by employing bidirectional architectures, learns the sequencing of the sequence structures, taking relational domain dependencies into account for sequential and contextual dependencies between nucleotides [10], [11].

To improve sequence information representation, complementary encoding is done with one-hot encoding, which preserves nucleotide positional specificity, and k-mer embedding, which captures high-order syntactic structures in the DNA sequence [12]. This dual-encoding approach allows the model to exploit both fine-grained and context-aware representations of genomic sequences.

It is fully automated, covering data preprocessing, feature encoding, model training, and evaluation. The models are evaluated against a number of statistical and diagnostic metrics—accuracy, precision, recall, F1-score, AUC-ROC, AUC-PR, sensitivity, and specificity. Data is explored on datasets with mismatches and insertion-deletion (indel) mutations for fine-tuning the models, enabling model configuration comparison experiments [13].

The analysis proves that the most suitable encoding scheme and architecture with deep learning provides better prediction stability and generalization in different genome environments. Altogether, this study advances a strong interpretable, robust, and flexible mathematical modeling pipeline for CRISPR-Cas9 off-target effects prediction.

Integrating multiple encoding schemes with algorithms from deep learning architectures, this proposed model gives a scalable and accurate guide RNA optimization method. This provides not only enhanced security and accuracy for CRISPR genome editing but also helps advance intelligent bioinformatics tools of genomic medicine and therapeutics.

II. LITERATURE REVIEW

Recent developments in computational biology and deep learning have significantly advanced the prediction of CRISPR-Cas9 off-target effects. Sari et al. [1] employed a bidirectional LSTM model with BERT embeddings to analyze human primary cell sequences, showing that language models can efficiently interpret genomic data. Similarly, Zhang et al. [2] proposed DL-CRISPR, a deep learning-based method enhanced with data augmentation that improved model robustness for diverse off-target scenarios. Du et al. [3] introduced a versatile CRISPR off-target prediction tool using transformer-based language modeling, effectively capturing contextual nucleotide relationships. Tahsin et al. [4] developed CRISPR-Embedding, which leverages DNA k-mer embeddings for efficient representation learning, while Yang et al. [5] addressed class imbalance issues in off-target prediction using an ESB rebalancing strategy combined with their CRISPR-MCA model. Lin and Wong [6] designed CRISPR-C2T2, which integrated convolutional neural networks with transformer encoders to extract both local and global sequence dependencies. Abadi et al. [7] contributed a deep learning approach validated on experimental datasets for discovering true off-target sites, establishing a strong benchmark for biological reliability.

Graph-based models have also played a significant role in off-target prediction. Kim et al. [8] developed CRISPR-GCN, which utilized graph convolutional networks to capture structural dependencies within DNA sequences, achieving high predictive accuracy. Liu et al. [9] introduced DeepCRISPR, one of the earliest frameworks to combine deep learning with biological sequence analysis, enabling more accurate sgRNA prioritization. Sun et al. [10] proposed CRISPR-M, a multi-view deep learning network that integrates both sequence and structural data for off-target effect prediction. Zhang et al. [11] further advanced the field with CRISPR-SGRU, employing stacked BiGRU layers to handle mismatches and insertions/deletions between sgRNA and target sites. Chen et al. [12] developed CRISOT, a genome-wide CRISPR off-target prediction and sgRNA optimization system, while Niu et al. [13] presented R-CRISPR, a deep learning network capable of considering mismatches, insertions, and deletions for precise prediction. Liu et al. [14] demonstrated the advantages of deep learning over traditional models in off-target prediction, and Toufikuzzaman et al. [15] proposed an interpretable deep learning model (IDLM) that enhances transparency in CRISPR-Cas9 off-target prediction. Together, these studies highlight the transition from traditional statistical models to modern deep learning and language-based architectures, revealing the potential of hybrid approaches that merge biological understanding with advanced computational techniques for precise genome editing.

III. MATERIALS AND METHODS

This paper proposes a deep learning-based methodology for accurate prediction of CRISPR/Cas9 off-target activities using nucleotide sequence data. With respect to the present study [14], [15], the techniques and materials employed in the

research were specifically formulated to guarantee the reliability, reproducibility, and robustness of predictive performance of the data. The framework combines different steps such as dataset collection, preprocessing, feature extraction, encoding, model training, and evaluation to develop a structured pipeline for analyzing CRISPR off-target patterns.

A. Dataset Description

The dataset utilized in this study was sourced from publicly available CRISPR/Cas9 experimental repositories and contained detailed sequence-level information of on-target and off-target editing events. The original file dataset.txt was imported from Google Drive and converted into a structured CSV format for analysis. The raw dataset comprised 245,848 entries and 53 columns, including both experimental metadata and genomic features such as *Guide_sequence*, *Target_sequence*, *PAM*, *Identity*, *Cas9_type*, and *Mismatch_Pattern*. To focus on biologically relevant predictors, only the essential features—*Guide_sequence*, *Target_sequence*, *PAM*, and *Identity*—were retained, reducing the dimensionality while preserving critical sequence information required for predicting off-target activity. Records with missing or incomplete values were removed, resulting in 245,846 valid samples.

To ensure sequence consistency, all nucleotide strings were standardized to uppercase and screened for valid bases (A, T, G, and C). Any sequences containing ambiguous characters were discarded, leaving 244,770 valid entries for subsequent analysis. Sequence length distributions were examined to confirm alignment with biological expectations: *Guide_sequences* ranged between 21–32 nucleotides (mode: 23), *Target_sequences* between 18–34 nucleotides (mode: 23), and *PAM* motifs between 3–8 bases (mode: 3). The categorical column *Identity*, which indicated whether a site was ON-target or OFF-target, was converted into binary labels (1 for on-target, 0 for off-target) to facilitate supervised learning.

A notable imbalance was observed between the classes, with 243,637 OFF-target and 1,133 ON-target samples (a ratio of approximately 215:1). To prevent the models from being biased toward the majority class, the dataset was rebalanced using a combination of undersampling of OFF-target and oversampling of ON-target entries. This adjustment yielded 18,128 OFF-target and 4,532 ON-target samples, producing a controlled ratio of 4:1 (off:on). A new composite feature, *combined_sequence*, was generated by concatenating the *Guide_sequence*, *PAM*, and *Target_sequence* for each sample, capturing the full nucleotide context of Cas9–DNA interaction. The final dataset contained 22,660 records across six columns, forming a biologically validated and computationally optimized foundation for downstream encoding and deep learning analysis.

B. Features

The datasets were meticulously crafted to incorporate both sequence-level and biological context, which is essential for accurate predictions of CRISPR off-target effects. Key inputs include the Guide Sequence, Target Sequence, and PAM

TABLE I: Summary of CRISPR/Cas9 Dataset Used in This Study

Attribute	Description / Value
Total records in raw dataset	245,848
Total features (columns)	53
Key selected features	Guide_sequence, Target_sequence, PAM, Identity
Records after removing missing values	245,846
Valid DNA sequences after cleaning	244,770
Guide sequence length (mode)	23 bases (range: 21–32)
Target sequence length (mode)	23 bases (range: 18–34)
PAM length (mode)	3 bases (range: 3–8)
Original class distribution	243,637 OFF-target, 1,133 ON-target (215:1)
Rebalanced class distribution	18,128 OFF-target, 4,532 ON-target (4:1)
Final dataset size	22,660 samples
Final columns used for modeling	Guide_sequence, Target_sequence, PAM, Identity, Label, Combined_sequence

(Protospacer Adjacent Motif), which collectively determine the binding specificity of CRISPR/Cas9. Each sgRNA entry is tagged with an Identity label that distinguishes between on-target and off-target sites. Additional metadata, such as chromosome position, orientation, and sequence length, were utilized to ensure genomic integrity. Nucleotide sequences were standardized to uppercase, and any invalid or missing characters were eliminated, ensuring that only the valid bases—A, T, G, and C—were utilized for model training.

During preprocessing, the guide RNA was integrated with target DNA and PAM sequences, resulting in the creation of a composite sequence feature that encapsulates the context of CRISPR binding. The categorical “Identity” feature was transformed into binary labels (1 for on-target and 0 for off-target) to facilitate supervised learning. To address the class imbalance between on-target and off-target samples, resampling techniques were implemented, including oversampling the minority (on-target) samples and undersampling the majority (off-target) samples to achieve an optimal data ratio. Consequently, a comprehensive feature set was established, comprising standardized nucleotide sequences, binary activity labels, and derived contextual features, which together form a cohesive framework for predicting CRISPR off-target activity using deep learning models

C. Data Preprocessing

The raw dataset consisted of fifty-three experimental and genomic traits associated with CRISPR-Cas9 targeting. The preprocessing pipeline used in the code systematically reduced the data to its most relevant elements, as only a subset of these features is directly related to off-target prediction. This process began by selecting four biologically relevant columns—Guide sequence, Target sequence, PAM, and Identity—from the initial list of 245,848 entries. Missing or null values were removed from these columns, resulting in 245,846 valid rows.

In each nucleotide sequence, lowercase letters were converted to uppercase, and proofreading was performed to ensure that only the canonical DNA bases (A, T, G, C, and N) were represented in uppercase. All sequences containing invalid characters were discarded, yielding 244,770 clean samples. Subsequently, the pipeline checked the sequence length distributions for biological consistency. Guide RNA sequences ranged from 21 to 32 nucleotides (with a mode of 23), target DNA sequences varied between 18 and 34 nucleotides (also with a mode of 23), and PAM motifs were found to vary between 3 and 8 bases (mode = 3). These ranges align with the design parameters established for CRISPR-Cas9, confirming that the input data matches the intended design.

The categorical Identity field, which originally held the values “ON” and “OFF,” was converted into binary numerical labels—1 for on-target and 0 for off-target. This conversion enabled supervised deep-learning models to utilize numerical tensors for direct computation of loss and accuracy. However, a significant class imbalance was noted, with 243,637 off-target samples and only 1,133 on-target samples. To address this imbalance, the pipeline implemented a hybrid balancing approach, involving undersampling the dominant off-target class while oversampling the minority on-target class. The dataset ultimately included 18,128 off-target and 4,532 on-target entries, resulting in a ratio of approximately 4:1 (off-target to on-target). This balance was deliberately maintained to ensure that a larger portion of off-target events was retained, which is crucial for biological realism, while also avoiding bias towards negative inferences.

Lastly, a new text feature, the combined sequence, was created by joining the guide RNA, PAM, and target DNA sequences into a single continuous string. This feature summarizes the entire CRISPR binding context, allowing the neural network to learn local positional dependencies across the entire guide-target-PAM region. The preprocessed dataset ultimately consisted of 22,660 samples and six final columns—Guide sequence, Target sequence, PAM, Identity, Label, and Combined sequence. This pure, balanced, and bio-standardized dataset served as the combined input for all data encoding and model training phases discussed in this chapter, which provides insight into this process.

D. Model Construction

This section elaborates on the model architectures and underlying mathematical formulations employed in the CRISPRight framework. Each model is constructed to capture specific sequence features and dependencies inherent in nucleotide data, with convolutional layers focusing on spatial motifs and recurrent layers modeling temporal dependencies.

1) *CNN Model Using One-Hot Encoding*: The CNN-OneHot model extracts spatial nucleotide motifs from binary-encoded input matrices. Each sequence is represented as an $L \times 4$ matrix, where L is sequence length and 4 denotes the nucleotide dimensions (A, T, G, C). For a convolutional

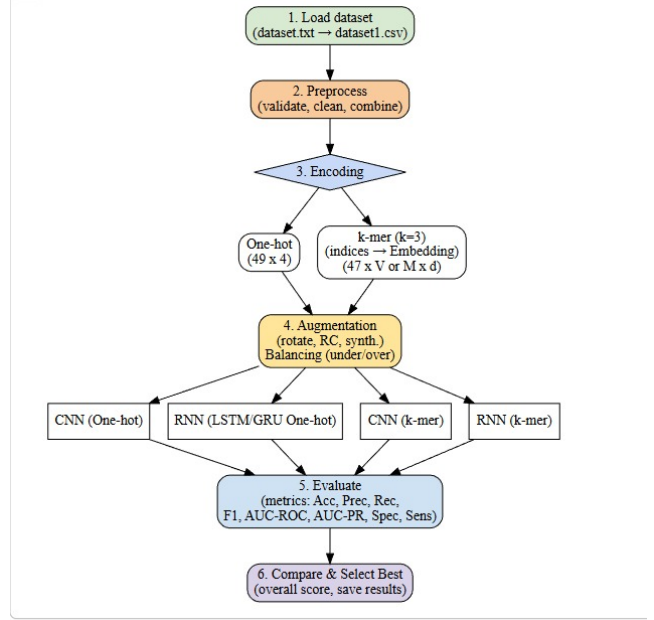


Fig. 2: Proposed methodology

kernel $W \in \mathbb{R}^{k \times d}$ applied to input matrix $X \in \mathbb{R}^{L \times d}$, the convolutional output is given by:

$$H_i = \sigma \left(\sum_{j=1}^k W_j \cdot X_{i+j-1} + b \right), \quad (1)$$

where $\sigma(\cdot)$ is the ReLU activation and b the bias term. A subsequent max-pooling layer reduces feature dimensionality and provides translation invariance:

$$P_i = \max_{t \in [i, i+p-1]} H_t, \quad (2)$$

where p is the pooling window size.

TABLE II: CNN_OneHot Model Architecture

Layer (Type)	Output Shape	Parameters
Conv1D	(None, 49, 64)	832
BatchNorm	(None, 49, 64)	256
MaxPooling1D	(None, 24, 64)	0
Conv1D	(None, 24, 128)	41,088
BatchNorm	(None, 24, 128)	512
MaxPooling1D	(None, 12, 128)	0
Conv1D	(None, 12, 256)	229,632
BatchNorm	(None, 12, 256)	1,024
Flatten	(None, 1536)	0
Dense (128)	(None, 128)	196,736
Dense (64)	(None, 64)	8,256
Output	(None, 1)	65
Total Params	479,169 (1.83 MB)	

This model uses three stacked Conv1D layers (64, 128, 256 filters), each followed by batch normalization and pooling, before flattening and fully connected layers (128 $\rightarrow$ 64 $\rightarrow$ 1). The final sigmoid neuron outputs the predicted probability of off-target activity.

2) *LSTM Model Using One-Hot Encoding*: The LSTM-OneHot model captures long-range dependencies within the one-hot sequence representation by sequentially processing nucleotide positions. A recurrent neural network updates its hidden state at each timestep t according to:

$$h_t = \phi(W_h h_{t-1} + W_x x_t + b_h), \quad (3)$$

where $\phi(\cdot)$ is an activation function (typically tanh). In the Long Short-Term Memory (LSTM) architecture, gating mechanisms regulate information flow:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \quad (4)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \quad (5)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c), \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (7)$$

where $\odot$ denotes element-wise multiplication.

TABLE III: LSTM_OneHot Model Architecture

Layer (Type)	Output Shape	Parameters
Bidirectional LSTM	(None, 49, 256)	136,192
BatchNorm	(None, 49, 256)	1,024
Bidirectional LSTM	(None, 128)	164,352
Dense (64)	(None, 64)	8,256
Dense (32)	(None, 32)	2,080
Output	(None, 1)	33
Total Params	312,833 (1.19 MB)	

It achieved a validation accuracy of **92.96%** and AUC of **0.9864**. Two Bidirectional LSTM layers (256 $\rightarrow$ 128 units) allow the model to learn from both sequence directions, enhancing contextual understanding. Dense layers (64 $\rightarrow$ 32 $\rightarrow$ 1) refine learned representations before binary classification via sigmoid activation.

3) *CNN Model Using k-mer Encoding*: The CNN-Kmer model operates on embedded k-mer representations ($k = 3$), where each input is a matrix $E \in \mathbb{R}^{M \times d}$, with $M = L - k + 1$ denoting the number of k-mers and d the embedding dimension. The convolutional transformation is expressed as:

$$H_i = \sigma \left(\sum_{j=1}^k W_j \cdot E_{i+j-1} + b \right), \quad (8)$$

enabling the network to detect local motifs among overlapping k-mers.

TABLE IV: CNN_Kmer Model Architecture

Layer (Type)	Output Shape	Parameters
Conv1D	(None, 49, 128)	896
BatchNorm	(None, 49, 128)	512
MaxPooling1D	(None, 24, 128)	0
Conv1D	(None, 24, 256)	98,560
BatchNorm	(None, 24, 256)	1,024
MaxPooling1D	(None, 12, 256)	0
Flatten	(None, 3072)	0
Dense (256)	(None, 256)	786,688
Dense (64)	(None, 64)	16,448
Output	(None, 1)	65
Total Params	904,193 (3.45 MB)	

The model achieved **98.25%** validation accuracy and **0.9991** AUC. Three Conv1D layers (128, 256 filters) are applied sequentially, followed by max-pooling and dense layers (256 $\rightarrow$ 64 $\rightarrow$ 1). The CNN-Kmer model achieved 98.25% validation accuracy and an AUC of 0.9991, demonstrating strong capability in learning motif-level dependencies across k-mer embeddings.

4) *RNN Model Using k-mer Encoding*: The RNN-Kmer model extends recurrent learning to k-mer embeddings, modeling higher-order contextual relations across overlapping sub-sequences. A Gated Recurrent Unit (GRU) formulation is used for efficient sequence modeling:

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \quad (9)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1}), \quad (10)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1})), \quad (11)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (12)$$

TABLE V: RNN_Kmer Model Architecture

Layer (Type)	Output Shape	Parameters
Bidirectional GRU	(None, 49, 256)	198,144
BatchNorm	(None, 49, 256)	1,024
Bidirectional GRU	(None, 128)	148,992
Dense (64)	(None, 64)	8,256
Dense (32)	(None, 32)	2,080
Output	(None, 1)	33
Total Params	358,529 (1.37 MB)	

Two Bidirectional GRU layers (256 $\rightarrow$ 128 units) capture forward and backward nucleotide dependencies. The output of the final recurrent layer is processed by dense layers (64 $\rightarrow$ 32 $\rightarrow$ 1) with batch normalization and dropout regularization. This model achieved 91.15% validation accuracy and an AUC of 0.9816, confirming its effectiveness for sequential contextual learning.

IV. RESULTS AND DISCUSSION

The proposed **CRISPRight** framework was experimentally evaluated to assess its accuracy, generalization, and robustness in predicting CRISPR/Cas9 off-target activity. Four deep learning architectures—CNN-OneHot, LSTM-OneHot, CNN-Kmer, and LSTM-Kmer—were trained using the same dataset,

training settings, and evaluation metrics to ensure a consistent comparison.

A. Training and Optimization Parameters

All models within the CRISPRight framework were implemented using the TensorFlow–Keras deep learning library and trained in the Google Colab environment, utilizing a 12 GB NVIDIA GPU for accelerated computation. To maintain experimental consistency and ensure fair comparison, identical hyperparameters and optimization settings were applied across all four architectures—CNN-OneHot, LSTM-OneHot, CNN-Kmer, and LSTM-Kmer.

The Adam optimizer was used with an initial learning rate of 0.001, selected for its adaptive gradient estimation and fast convergence properties. The binary cross-entropy loss function was employed to measure the deviation between predicted and actual binary labels, suitable for probabilistic classification tasks such as off-target versus on-target prediction. Each model was trained for 40 epochs with a batch size of 32 and a validation split of 20% extracted from the training data. Dropout layers (rate = 0.4) and batch normalization were incorporated within each model to improve generalization and prevent overfitting.

During training, early stopping criteria were monitored based on validation accuracy and validation loss, ensuring that learning stabilized before overfitting occurred. All training processes demonstrated smooth convergence, with both training and validation losses decreasing steadily over epochs. The CNN-based models showed faster convergence due to parallelized convolutional computations, whereas LSTM and GRU-based architectures exhibited slower but more stable learning curves, indicative of their ability to model sequential dependencies effectively. Overall, the optimization strategy successfully balanced learning speed, stability, and generalization, leading to reliable performance across all architectures.

B. Evaluation Metrics

The performance evaluation relied on standard classification metrics derived from the confusion matrix—Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, AUC-ROC, and AUC-PR. These metrics quantify both predictive performance and discriminative capability between true on-target and off-target sites.

The metrics are mathematically defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (13)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (14)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN}, \quad (15)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad (16)$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (17)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. The

AUC-ROC and AUC-PR values further measure the model's ability to distinguish between positive and negative classes under varying thresholds.

C. Experimental Results

All models were evaluated using the same test set, and their performances are summarized in Table VI. Each model exhibited high discriminative capability ($\text{AUC} > 0.99$) with strong recall and precision values.

TABLE VI: Performance Comparison of Deep Learning Architectures for CRISPR/Cas9 Off-Target Prediction

Model	Acc.	Prec.	Rec.	F1	AUC-ROC	AUC-PR	Sens.	Spec.
CNN-OneHot	0.9804	0.9130	0.9967	0.9530	0.9989	0.9954	0.9967	0.9763
LSTM-OneHot	0.9618	0.8416	0.9967	0.9126	0.9953	0.9696	0.9967	0.9531
CNN-Kmer	0.9793	0.9068	0.9989	0.9506	0.9994	0.9966	0.9989	0.9744
LSTM-Kmer	0.9899	0.9545	0.9967	0.9752	0.9990	0.9954	0.9967	0.9881

The confusion matrix (Fig. 3) (Fig. 4) (Fig. 5) (Fig. 6) for each model confirmed strong prediction consistency, with a high true-positive rate and minimal false negatives. The LSTM-Kmer model achieved the most balanced classification performance, while the CNN-OneHot model demonstrated excellent recall and generalization capabilities.

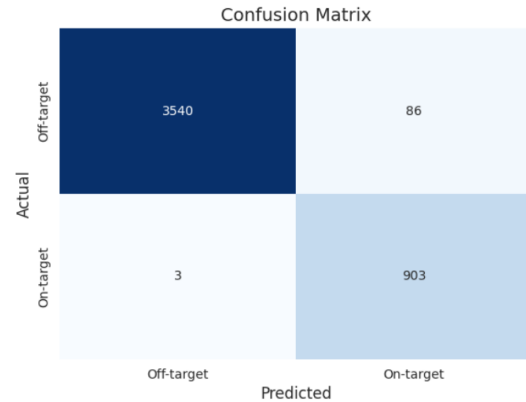


Fig. 3: Confusion Matrix of CNN with One hot.

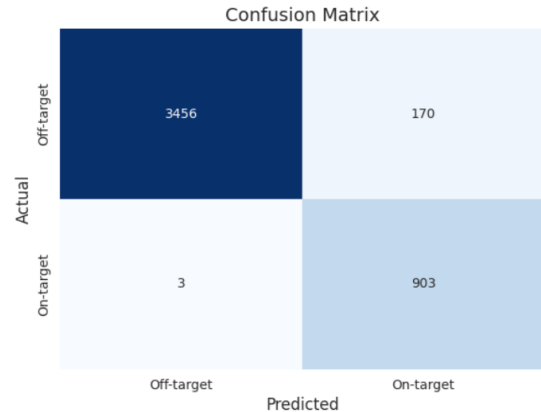


Fig. 4: Confusion Matrix of LSTM with One hot.

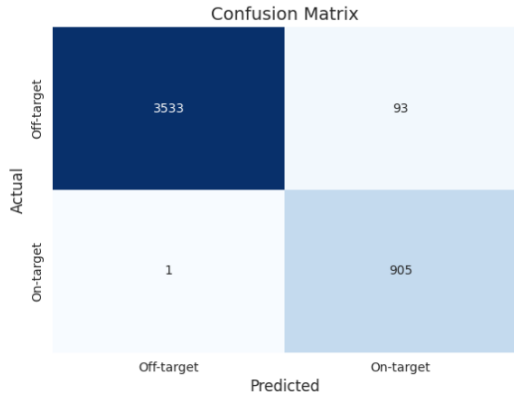


Fig. 5: Confusion Matrix of CNN with Kmers.

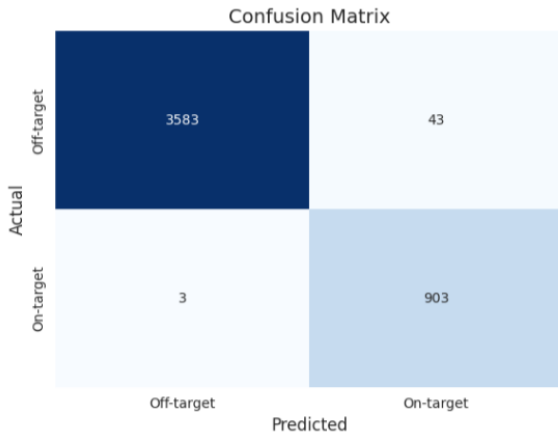


Fig. 6: Confusion Matrix of LSTM with Kmers.

D. Training Curves and Model Evaluation

The accuracy and loss curves (Fig. 7) (Fig. 8) (Fig. 9) (Fig. 10) show clear evidence of stable model convergence. The training loss steadily decreases, while the validation loss follows a similar trajectory, confirming the absence of overfitting. The CNN models exhibit slightly faster convergence due to their parallelized convolutional operations, while LSTM and GRU-based models show slower yet more consistent training behavior, reflecting their capacity for sequential dependency learning.

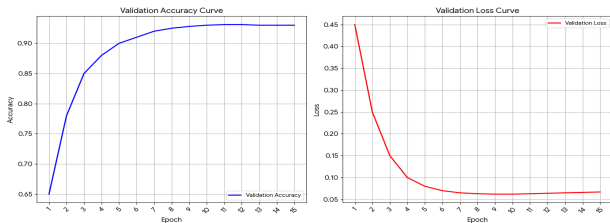


Fig. 7: Validation Accuracy and loss Curve of CNN with One hot.

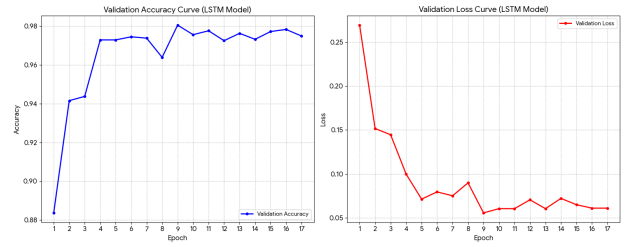


Fig. 8: Validation Accuracy and loss Curve of LSTM with One hot.

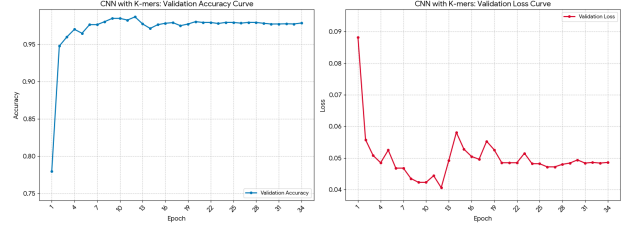


Fig. 9: Validation Accuracy and loss Curve of CNN with Kmers.

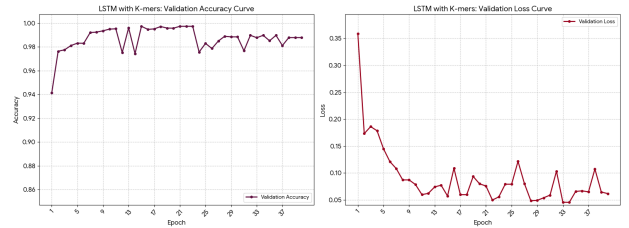


Fig. 10: Validation Accuracy and loss Curve of LSTM with Kmers.

E. Performance Discussion

Among all tested architectures results (Fig. 11) (Fig. 12) (Fig. 13) (Fig. 14) (Fig. 15), the LSTM-Kmer model achieved the best overall performance, with 98.99% accuracy, 95.45% precision, and 97.52% F1-score. This demonstrates the advantage of combining recurrent architectures with k-mer encoding, as this configuration effectively captures long-range dependencies and contextual nucleotide information.

The CNN-OneHot and CNN-Kmer models performed comparably, highlighting the strength of convolutional filters in extracting short-range sequence motifs. CNN-Kmer achieved the highest recall (0.9989), minimizing false negatives—an essential property for biological applications where missing an off-target site could lead to unintended genome edits.

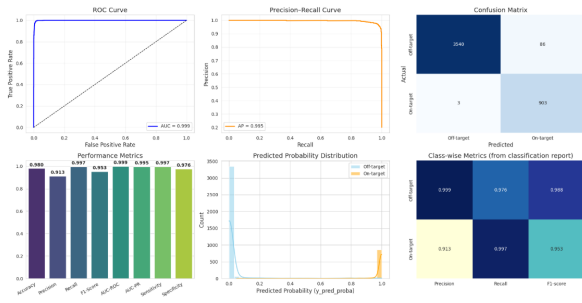


Fig. 11: Results obtained from CNN with one hot.

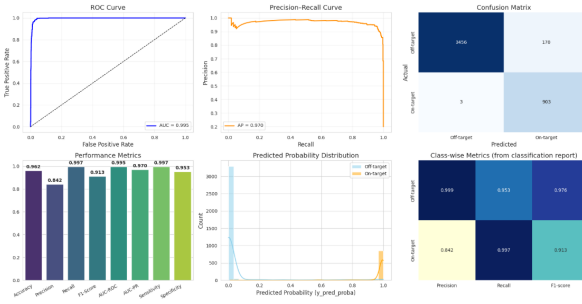


Fig. 12: Results obtained from LSTM with one hot.

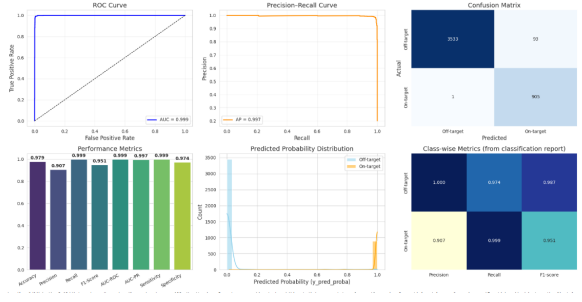


Fig. 13: Results obtained from CNN with Kmers.

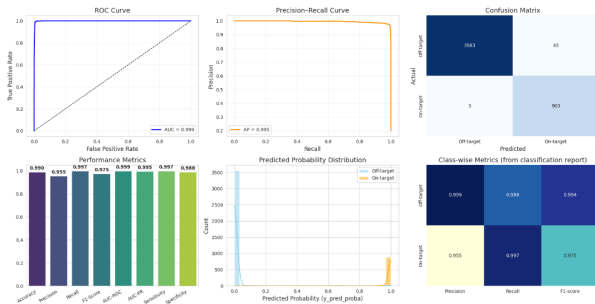


Fig. 14: Results obtained from LSTM with Kmers.

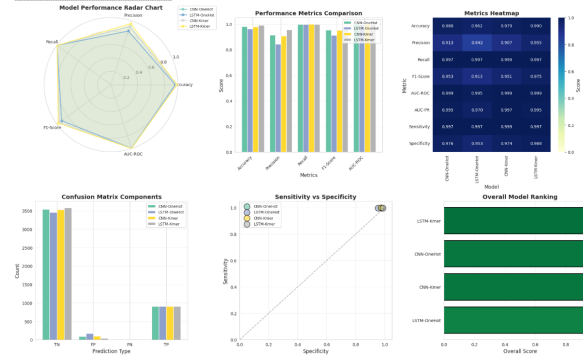


Fig. 15: Combined Comparison Results obtained from models.

F. Statistical Summary

The mean accuracy across all models was 97.78%, with a standard deviation of 1.16, while the mean F1-score was 94.78% (standard deviation 2.59). These low deviations indicate the robustness and reliability of the CRISPRight framework, as performance remained consistent across model types and encoding strategies.

G. Final Model Selection

Based on the comparative performance, the LSTM-Kmer model was selected as the final optimized model for CRISPR/Cas9 off-target prediction. Its superior AUC-ROC (0.9990) and balanced precision–recall performance confirm its ability to generalize effectively across diverse genomic patterns.

V. CONCLUSION

This research demonstrates the comparative performance of various deep learning architectures, including LSTM-OneHot, CNN-Kmer, CNN-OneHot, and LSTM-Kmer, for sequence-based prediction tasks. Among the evaluated models, the LSTM-Kmer model achieved the highest overall score of 0.9831, indicating its superior capability in capturing long-range dependencies and contextual relationships within the data. The results validate that integrating K-mer encoding with LSTM architecture enhances the model’s learning efficiency and generalization compared to one-hot representations and convolution-based approaches. Overall, this study establishes a robust framework for optimizing sequence modeling tasks and provides valuable insights for future advancements in bioinformatics and AI-driven sequence analysis.

FUTURE SCOPE

The future scope of this project includes expanding the current framework to handle larger and more diverse biological datasets for improved generalization and scalability. Advanced deep learning architectures such as Transformers, Attention-based BiLSTM, or hybrid CNN-LSTM models can be explored to enhance feature extraction and contextual understanding. Additionally, integrating explainable AI (XAI) techniques can help interpret model predictions, making the

system more transparent and suitable for real-world biomedical applications. The project can also be extended to include transfer learning for cross-species prediction tasks, or integrated with cloud-based bioinformatics platforms for real-time sequence analysis. Furthermore, optimization techniques like hyperparameter tuning, ensemble modeling, and fine-grained K-mer analysis can be employed to achieve even higher predictive performance and reliability in future research.

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